VENKATA SAI GANESH MANDA

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Target Roles: SDE | AI/ML Engineer | Robotics SDE | Data Scientist/Engineer (Spring, Summer 2027)

SUMMARY

Master's student in Robotics and Autonomous Systems (AI) at Arizona State University (GPA 4.0) with hands-on experience building scalable distributed systems, LLM-powered multi-agent pipelines, production RAG applications, and robotics software integrating computer vision, motion planning, and autonomous control. Proven track record modifying open-source frameworks, writing automated test scripts, analyzing large datasets, and authoring technical documentation. Proficient in Python, C++, SQL, R, OOP, data structures and algorithms, CI/CD, and Agile SDLC. Passionate about owning the full software lifecycle from design through deployment across AI, robotics, and data engineering domains.

EDUCATION

M.S. Robotics and Autonomous Systems (AI), Arizona State University **Aug 2025 – May 2027**
SCAI, Ira A. Fulton Schools of Engineering, Tempe, AZ | GPA: 4.0/4.0
Relevant Coursework: Applied Linear Algebra for Engineers, Robotic Systems I, Agentic AI, Statistical Machine Learning

B.Tech. Artificial Intelligence and Data Science, IIITDM Kurnool **Sep 2020 – Jun 2024**
Exchange student, University of Agder (UiA), Grimstad, Norway | Jan 2024 – Jun 2024

TECHNICAL SKILLS AND CERTIFICATIONS

Languages: Python, C++, C, R, SQL, JavaScript/TypeScript (basic)
Software Eng: OOP, Data Structures and Algorithms, Distributed Systems, Design Patterns, Agile/Scrum, CI/CD, Git (CLI), Code Review, Debugging, SDLC
AI/ML: PyTorch, TensorFlow, Scikit-learn, LangChain, RAG, Multi-Agent Architectures, OpenAI API, DeepSeek API, Gemini API, Prompt Engineering
Robotics and CV: ROS 2, OpenCV, Computer Vision, Homography, Perspective Transforms, Affine Calibration, Motion Planning, Path Planning
Data Science: EDA, Statistical Modeling, Data Wrangling, Predictive Modeling, A/B Testing, Causal Inference, NumPy, Pandas, Hadoop, Spark
Testing and QA: Unit Testing, Smoke Testing, Pytest, Functional Testing, Test Automation, Defect Tracking, Validation and Verification
Cloud and Infra: AWS (ECS Fargate, ALB, S3, IAM), Azure AI Studio, ChromaDB, REST APIs, FastAPI, SLURM, HPC, GitHub Actions, Docker, Microservices
Certifications: ML Specialization (Coursera/Stanford); Big Data and Spark (EdX)

PROJECTS

CodeLens: RAG-Based Codebase Intelligence API | Independent Project **May 2026 – Present**
- Shipped production RAG API (hybrid BM25 + semantic retrieval, cross-encoder reranking, ChromaDB), deployed via Docker on AWS ECS Fargate.
- Built GitHub Actions CI/CD gating merges on quality checks; cut full repo reindex from 3.5 hours to minutes.
- Cut eval cost 96% to $0.62/run (saving ~87K GPT-4o tokens/run) while holding ~80% faithfulness and answer-relevancy. https://github.com/mvsg2/CodeLens | Demo: https://youtu.be/zm7Cu2Lo5Ww

Tau-Bench: Multi-Agent LLM Infrastructure | Arizona State University **Jan 2026 – May 2026**
- Modified 11 vLLM source files to enable distributed model serving on an HPC cluster across 825 runs (165 tasks x 5 trials).
- Wrote Pytest smoke/unit tests for critic-agent eval; co-authored 115 SLURM scripts for fault-tolerant distributed runs across Qwen3 models (4B-32B).
- Root-caused 3,677 LLM agent failures; redesigned architecture around Planner-Critic-Executor to fix the dominant failure mode. https://github.com/mvsg2/tau-bench-CSE598-repo

Autonomous Maze-Solving Robot | Arizona State University **Aug 2025 – Dec 2025**
- Built Python perception-planning-control pipeline (OpenCV) converting raw camera frames into structured occupancy grids.
- Used Gemini 2.5 Flash tool calls for entrance/exit detection plus optimization-based planning for minimum-length waypoint sequences.
- Achieved 100% success across all tested 4x4 maze configs via 6-parameter affine least-squares pixel-to-robot calibration, regardless of camera orientation. https://github.com/mvsg2/RAS-545-Midterm-Project | Demo: https://www.youtube.com/watch?v=QzS8yIfglXg

LLM-Controlled Robotic Manipulation | Arizona State University **Aug 2025 – Dec 2025**
- Architected OOP agentic pipeline (Gemini API + ROS 2 + OpenCV) letting a robot arm do pick-and-place from natural language commands.
- Built Python tool-calling interface for pixel-to-world transforms with clean separation across vision, planning, and execution.
- Validated end-to-end autonomy with zero manual intervention across all evaluated object placements; published open-source with docs. https://github.com/mvsg2/RAS-545-Final-Project | Demo: https://youtu.be/sKL3z0AOqHU

Harmful Brain Activity Classification, EEG Pipeline | University of Agder **Jan 2024 – Jun 2024**
- Trained ResNet/EfficientNet/MobileNet in PyTorch (transfer learning, LR scheduling) on HMS spectrogram data; hit 71% validation accuracy.
- EfficientNet beat ResNet/MobileNet by approximately 5% in statistical benchmark.
- Applied adversarial learning and pseudo-labeling in bachelor's thesis; approximately 10% accuracy gain over baselines on Office-Home and DomainNet.

PROFESSIONAL EXPERIENCE

AI Intern | Semine AS, Kristiansand, Norway **Jan 2024 – Jun 2024**
- Designed and deployed document-intelligence pipeline on Azure AI Studio (OCR + REST field extraction); cut manual processing time 30% in production.
- Debugged data-quality defects via structured EDA over 300+ invoice records in a SQL Server-backed Python codebase.
- Fine-tuned document models across 200+ foreign-language invoice types; boosted extraction accuracy 15% over baseline OCR.